



Biomechanics of human movement 5 ECTS Group Report

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Contributions:

Imindu Dissanayaka: IMU signal analysis (section 2 to 4), section 9 (Product Concept).

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Both members: Introduction, Section 8 (Group challenge).

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1. Introduction

Laboratory measurements are a fundamental tool in biomechanics research, allowing researchers to capture and quantify human movement using objective sensor data. In this experiment, wearable sensors were used to record movement signals during squat exercises, enabling analysis of both kinematic and neuromuscular characteristics. Squats are a common functional movement used in sports, rehabilitation, and everyday life, making them a relevant subject for biomechanical study. By analyzing signals from multiple participants, it is possible to identify general movement patterns as well as individual variability.

In this experiment each participant did the following movement sequence: Jump, rest (30 seconds), 5 slow squats, rest (30 seconds), 5 fast squats, jump, rest (30 seconds). The data from each participant was combined and during this signal analysis it is analyzed as group data. For each part of this report a figure from the most suitable participant is used but analyses are made on a group level.

Each data set includes IMU signal from thigh with Z-axis accelerometer and gyroscope signal, IMU signal from calf with Z-axis accelerometer and gyroscope and EMG data. During the analysis first data quality is evaluated, then IMU signals from thigh and calf are compared to each other. After that gyroscope signals are analyzed, and analysis of EMG signals is done. The results are interpreted as what they really mean in terms of biomechanics. The aim of this group work is to do signal analysis for bigger dataset that consists of multiple subjects and to analyze the data biomechanically. Purpose is to connect signals to movement and to learn how to handle group data.

Through this task, we learned how raw biomechanical signals relate to different phases of human movement and how sensor placement affects the characteristics of the recorded signal. We also gained practical experience in group-level data processing, including signal quality evaluation, gyroscope-based repetition detection, EMG rectification and envelope extraction, and statistical comparison between movement conditions. Most importantly, this task emphasized that signal analysis is not only about processing data, but about understanding what the signals mean in the context of real human movement.

2. Data quality screening

Before conducting group-level analysis, all 12 datasets were visually inspected to identify and exclude recordings with insufficient signal quality. This screening step is essential because poor-quality data can distort group-level results and lead to incorrect biomechanical interpretations. Since the gyroscope signal is the primary signal used for repetition detection and kinematic analysis, it was used as the main basis for the quality evaluation.

For a dataset to be considered acceptable, the thigh gyroscope Z-axis signal was expected to show a clear and interpretable movement structure consistent with the measurement protocol. Specifically, an acceptable signal should display two distinct activity periods separated by a rest period: the first corresponding to 5 slow squats and the second to 5 fast squats. Within each activity period, the signal should show regular oscillating peaks reflecting the angular rotation of the thigh during the squat movement — descending into the squat and returning to standing. The slow squat period was expected to show lower-amplitude, more widely spaced peaks, while the fast squat period was expected to show higher-amplitude peaks occurring more closely together. Outside these activity windows, the signal should remain relatively flat, reflecting the rest periods. Signals that did not conform to this expected pattern were flagged for exclusion

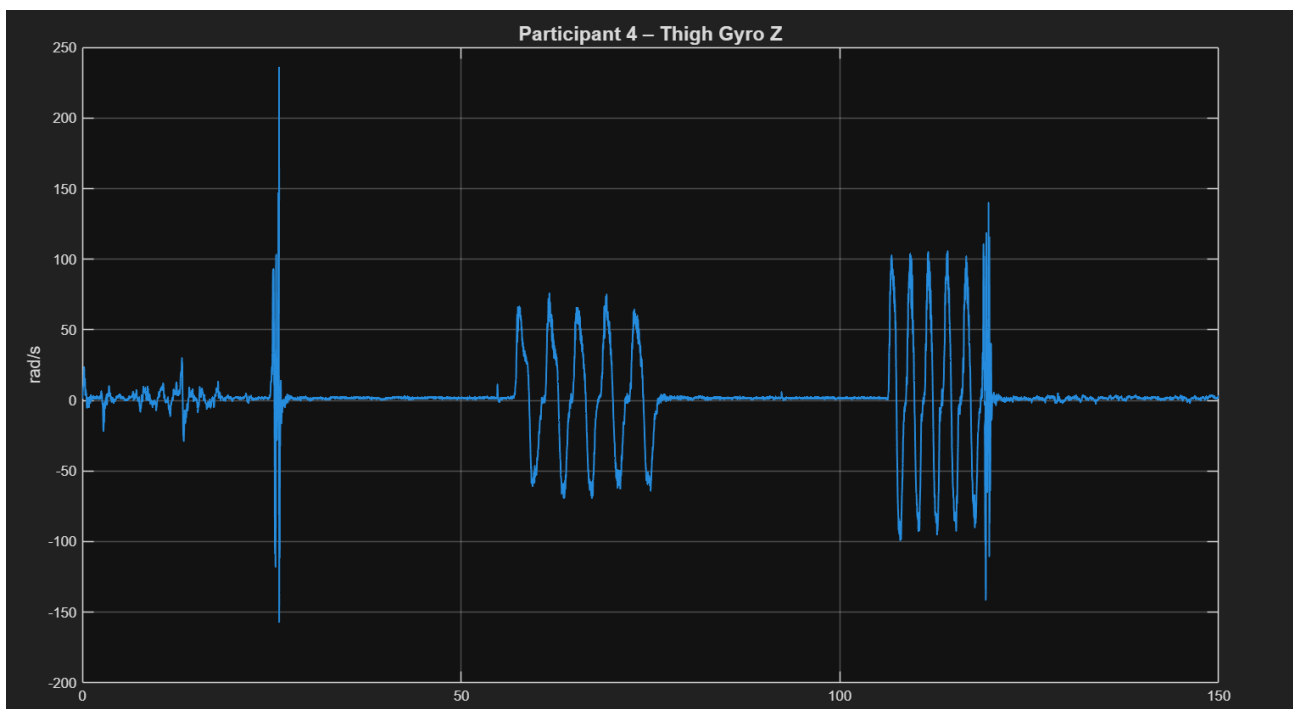


Figure 1: Correct Thigh gyroscope Z-axis signal (Participant 4)

Caption: Correct Thigh gyroscope Z-axis signal of Participant 4 showing clear squat phases, with sharp peaks

The full 150-second recording was reviewed for each participant, looking for the presence of clearly identifiable squat phases, appropriate signal amplitude, and absence of excessive noise or movement artefacts.

Following visual inspection, participants 5 and 12 were identified as the two weakest datasets and were excluded from all further analysis. When a participant was excluded, all corresponding signals were removed simultaneously — thigh gyroscope, calf gyroscope, thigh accelerometer, calf accelerometer, and EMG — ensuring consistency across the entire analysis pipeline. After exclusion, the final group consisted of $N = 10$ participants.

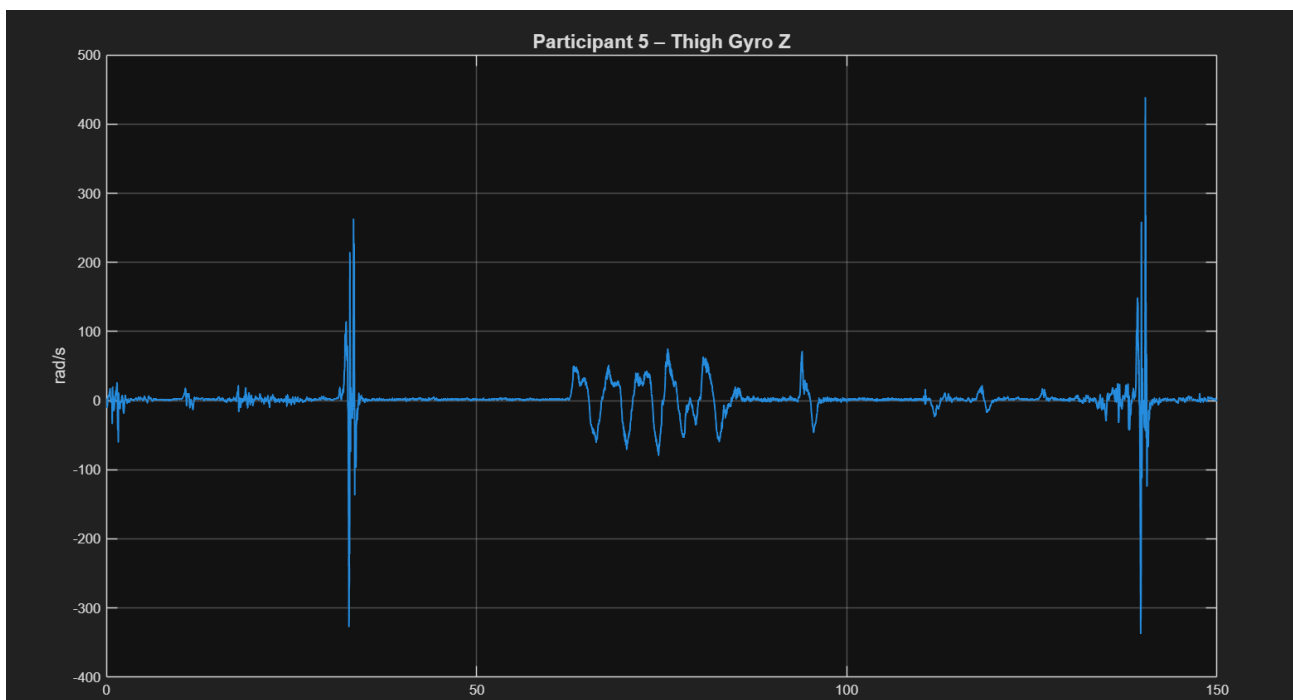


Figure 2: *Thigh gyroscope signal of Participant 5*

Caption: Thigh gyroscope Z-axis signal of Participant 5 showing inconsistent squat phases, irregular peak patterns, and sharp artefact spikes. This participant was excluded due to unclear movement phases and inconsistent signal patterns.

Participant 5 was excluded due to inconsistent and unclear movement phases. As seen in Figure 2, the slow squat period around 50–90 seconds shows irregular oscillations that do not follow a consistent repetition pattern. The peaks vary greatly in amplitude and shape, making it impossible to reliably identify individual squat repetitions. Additionally, the signal contains sharp isolated spike artefacts — notably around 25 seconds and again near 135–140 seconds — that are not consistent with normal squat movement and likely represent sensor movement or cable artefacts. The fast squat period is also difficult to distinguish clearly from noise. Due to these issues, automated peak detection would produce unreliable results for this participant.

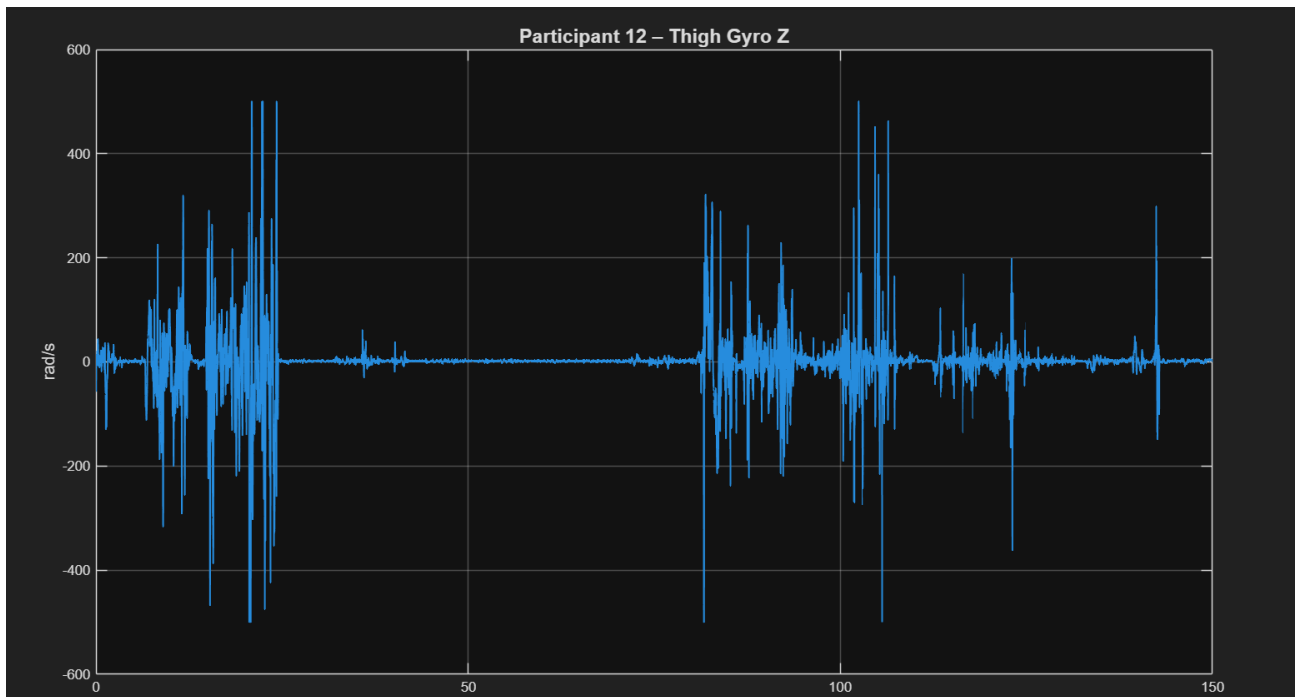


Figure 3: *Thigh gyroscope signal of Participant 12*

Caption: Thigh gyroscope Z-axis signal of Participant 12 showing excessive noise and high-amplitude artefacts throughout the recording. This participant was excluded due to severe signal contamination that makes movement phase identification impossible.

Participant 12 was excluded due to excessive noise and artefacts throughout the entire recording. As seen in Figure 3, the signal shows extremely high-amplitude irregular spikes reaching up to ± 500 rad/s, which far exceed the expected physiological range for squat angular velocity. These artefacts appear across both the supposed activity periods and rest periods, making it impossible to distinguish genuine squat movements from noise. There is no identifiable regular oscillation pattern that would correspond to slow or fast squats. The signal is dominated by large erratic bursts that are consistent with severe sensor malfunction, loose attachment, or significant external interference. Including this dataset would heavily distort group-level statistics, particularly peak angular velocity and repetition duration calculations.

3. IMU comparison (thigh vs calf)

To understand how different body segments respond to squat movement, the accelerometer and gyroscope signals from both the thigh and calf IMUs were plotted and compared for all 10 participants. This comparison helps to understand how motion is transmitted through the lower limb during squatting and why signal characteristics differ depending on sensor placement. For detailed illustration, Participant 4 was selected as a representative example, as their signals clearly show the expected movement structure and allow meaningful comparison between segments.

Overall observations

At a general level, both the thigh and calf signals show clear activity during the squat periods and relative silence during the rest periods, confirming that the IMU sensors successfully captured the movement. However, the nature and quality of the signals differ considerably between the two segments. The thigh signals — both gyroscope and accelerometer — show well-defined, rhythmic patterns during the slow and fast squat periods, closely reflecting the controlled rotational and linear motion of the thigh. The calf signals, by contrast, appear considerably weaker and less structured during the squat phases, with the movement pattern being much harder to identify visually.

Differences observed in Participant 4

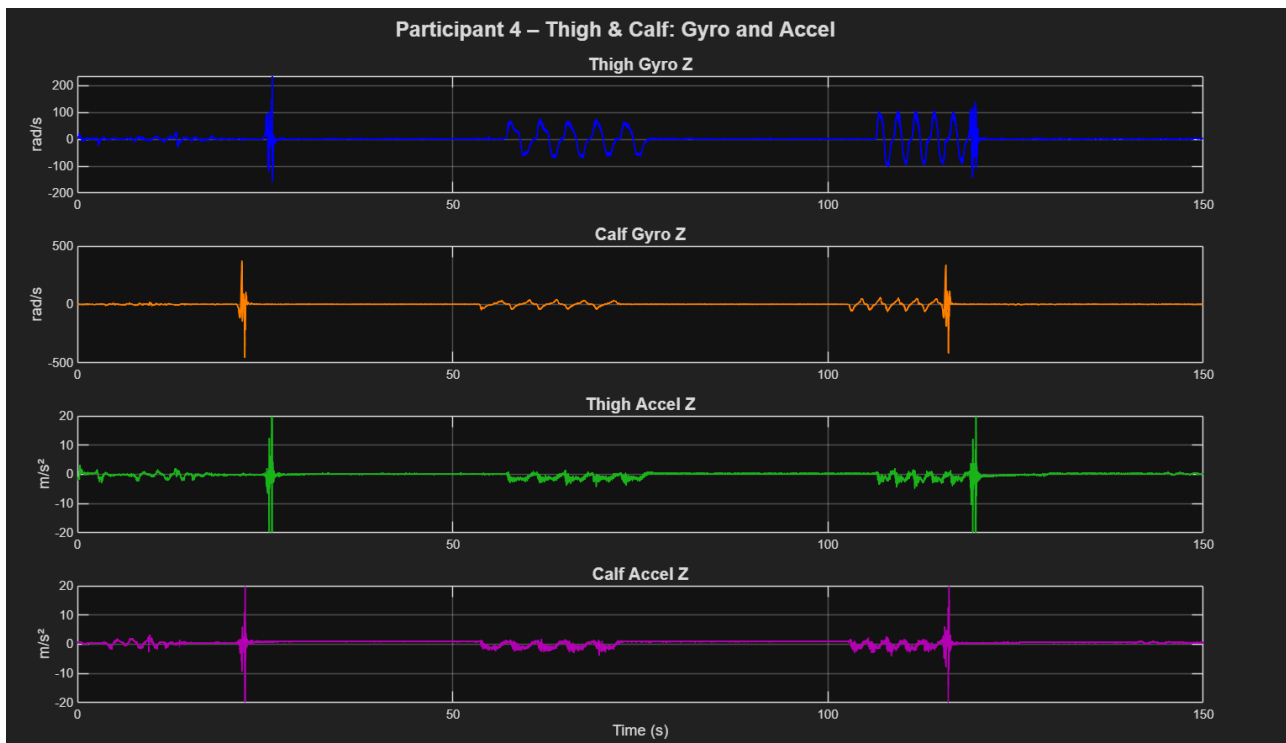


Figure 4: Participant 4 — Thigh and Calf Gyroscope and Accelerometer signals

Caption: Thigh gyroscope Z (blue), calf gyroscope Z (orange), thigh accelerometer Z (green), and calf accelerometer Z (pink) signals for Participant 4 over the full 150-second recording. Clear squat activity periods are visible in the thigh signals between approximately 50–90 s (slow squats) and 100–130 s (fast squats), while the calf signals show minimal movement-related activity during squatting.

Looking at Figure 4 (Participant 4), the differences between segments become very clear. The thigh gyroscope Z signal (blue) shows smooth, consistent oscillations during both squat periods — approximately 50–90 seconds for slow squats and 100–130 seconds for fast squats — with peak amplitudes reaching around ± 100 rad/s. The oscillations are regular and clearly reflect the repeated flexion and extension of the hip and knee as the thigh rotates during each squat repetition. Individual squat cycles are easily distinguishable, and the difference in rhythm between the slow and fast conditions is clearly visible.

The calf gyroscope Z signal (orange) shows only very small, low-amplitude fluctuations close to zero during the squat periods, making individual repetitions difficult or impossible to identify. The most prominent features in the calf gyroscope are two large spike artefacts near 25 seconds and 115 seconds, which coincide with the jump events rather than the squats. This indicates that the calf undergoes very little rotational movement during squatting compared to the thigh, as the calf segment remains relatively upright and stable while the thigh rotates considerably at the hip and knee joints.

The thigh accelerometer Z signal (green) shows recognizable rhythmic fluctuations during the squat periods corresponding to each repetition, along with two sharp impact spikes from the jumps. The calf accelerometer Z signal (pink) similarly shows large jump-related spikes but very limited and noisy activity during the squat periods, reflecting that the calf experiences mainly gravitational and postural acceleration rather than dynamic movement-related acceleration during squats.

Differences in accelerometer signals between segments

The accelerometer measures linear acceleration, which includes both movement-related dynamic acceleration and the constant gravitational component. During a squat, the thigh undergoes significant angular displacement as the knee flexes and extends, causing the orientation of the thigh IMU relative to gravity to change substantially. This produces a clear and measurable variation in the Z-axis accelerometer signal throughout each repetition. The calf, on the other hand, remains much closer to vertical throughout the squat — particularly in a standard squat where the calf maintains a more

upright position. Because the calf IMU orientation changes much less relative to gravity, the Z-axis accelerometer signal from the calf captures very little of the actual squat movement and instead reflects mostly static gravitational loading with small postural fluctuations. This fundamental difference in segmental kinematics explains why the thigh accelerometer produces a far more informative and movement-representative signal than the calf accelerometer during squatting.

Suitability of gyroscope for repetition detection

The gyroscope measures angular velocity directly, making it closely related to the rotational motion occurring at the joints during movement. During a squat, the thigh rotates in a consistent and predictable pattern with each repetition — flexing downward and extending back upward — producing a clear biphasic angular velocity signal. Unlike the accelerometer, the gyroscope is not affected by gravitational orientation, meaning its output reflects only actual rotational movement without the gravitational offset that complicates accelerometer-based analysis. This makes the gyroscope signal cleaner, more consistent across repetitions, and far more suitable for automated peak detection and repetition counting. As clearly seen in Participant 4's thigh gyroscope, each squat repetition produces a distinct, well-separated peak that can be reliably identified, whereas the accelerometer signal requires more complex processing to extract equivalent information.

4. Gyroscope

Repetition detection method Repetition detection method

Repetition detection was performed using the thigh gyroscope Z-axis signal as the primary source, as it provides the clearest and most consistent representation of squat movement among all available signals. The MATLAB findpeaks function was applied to detect positive peaks in the signal, which correspond to the upward phase of each squat repetition — the moment when the thigh reaches its maximum angular velocity while returning to the standing position. Two parameters were set to control detection quality: a minimum peak height threshold of 40 rad/s for the thigh and 16 rad/s for the calf, and a minimum peak distance of 200 samples (approximately 1.56 seconds at 128 Hz) to prevent multiple detections within a single repetition. The same approach was applied to the calf gyroscope for comparison purposes.

Detection results and challenges

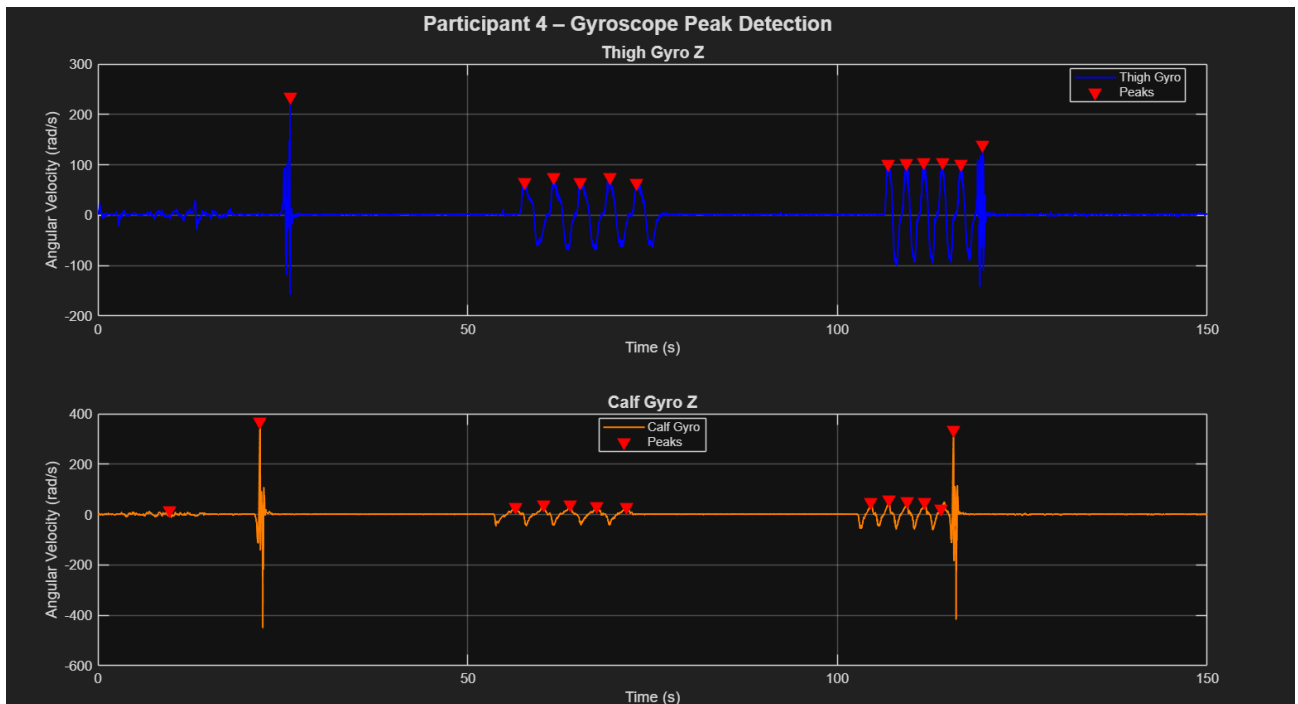


Figure 5: Participant 4 — Gyroscope Peak Detection, Thigh and Calf

Caption: Thigh gyroscope Z (blue, top) and calf gyroscope Z (orange, bottom) signals for Participant 4 with detected peaks marked by red downward triangles. Five slow squat peaks are visible between approximately 50–90 s and five fast squat peaks between approximately 100–130 s. Large artefact spikes near 25 s and 115 s correspond to jump events.

As seen in Figure 5 for Participant 4, the thigh gyroscope (blue) shows clearly identifiable peaks during both squat periods. Five peaks are detected in the slow squat period (approximately 50–90 seconds) and five in the fast squat period (approximately 100–130 seconds), correctly reflecting the protocol. The thigh signal produces reliable and well-separated peaks that align naturally with the squat repetitions, making it the most suitable signal for automated detection.

The calf gyroscope (orange) also shows detectable activity during squat periods, but the signal is considerably weaker and less consistent. While peaks are detected in the calf signal, their amplitudes are much lower and their shape less distinct compared to the thigh. The calf signal is more susceptible to false peak detection because small postural fluctuations and noise can reach the low detection threshold of 16 rad/s. Furthermore, during fast squats where repetitions occur in rapid succession, the minimum peak distance constraint of 200 samples occasionally causes some genuine peaks to be missed, as consecutive squats may occur closer together than this threshold allows. These limitations confirm that the thigh gyroscope is a more reliable source for repetition detection than the calf gyroscope.

Additionally, both signals contain large artefact spikes associated with the jump events at approximately 25 seconds and 115 seconds. These spikes far exceed the squat-related peak amplitudes

and can be falsely detected as peaks if the detection parameters are not carefully tuned. In the current implementation, the minimum peak distance parameter helps to partially suppress these artefacts, but visual verification of the detection results was still necessary to confirm correct identification of squat repetitions.

At below marked calculation thigh gyroscope of all participants:

Table I. *Thigh gyroscope calculations of all participants.*

Participant	Slow Squat				Fast Squat			
	Number of Repetitions	Peak Angular velocity	Minimum Angular velocity	Duration	Number Repetitions	Peak Angular Velocity	Minimum Angular Velocity	Duration
1	5	69.95	54.24	20.13	5	241.79	132.46	5.64
2	5	100.96	86.52	15.19	5	142.79	130.29	9.64
3	5	74.49	59.34	25.18	5	127.48	114.77	11.61
4	5	64.17	75.77	18.86	5	114.23	102.10	12.17
6	5	60.21	48.15	21.77	5	116.96	111.91	8.7
7	5	130.87	96.29	9.70	5	136.41	117.31	8.6
8	5	50.87	40.37	16.86	5	75.25	58.47	11.5
9	5	128.15	88.67	8.73	5	221.05	114.61	7.64
10	5	93.33	87.94	12.61	5	261.48	215.65	1.31
11	5	104.10	80.07	14.69	5	142.75	125.79	8.08

Understanding the variables

The table presents four gyroscope-derived variables for each participant across both squat conditions. Peak angular velocity represents the maximum rotational speed reached by the thigh during the upward phase of the squat — the moment of greatest rotational power output at the hip and knee joints. A higher peak angular velocity indicates that the participant moved more explosively and generated greater joint rotational power during that phase. Minimum angular velocity reflects the lowest point of angular speed within the squat cycle, corresponding to the brief transition moment at the bottom of the squat where the thigh momentarily decelerates before reversing direction. The difference between peak and minimum angular velocity therefore represents the dynamic range of the thigh rotation — a larger difference indicates a more pronounced and controlled squat movement with a clear deceleration and reacceleration pattern, while a smaller difference suggests a more continuous or less distinct movement. Repetition duration, expressed in seconds, describes how long each squat cycle took on average and directly reflects movement tempo. Longer duration corresponds to slower, more controlled squats requiring sustained muscular effort throughout the range of motion, while shorter duration reflects faster, more explosive movement with less time under tension per repetition.

Group-level observations

Across all 10 participants, a consistent pattern emerges: fast squats produced substantially higher peak angular velocities and considerably shorter repetition durations compared to slow squats. This is expected from the protocol and confirms that the gyroscope successfully captured the intended difference in movement tempo between the two conditions. However, the magnitude of these differences varies considerably between individuals, revealing interesting biomechanical differences in how participants executed the task.

Individual participant discussion

Participant 1 shows one of the most dramatic differences between conditions, with peak angular velocity increasing from a relatively modest slow squat value to an exceptionally high fast squat value, and repetition duration dropping sharply. This suggests that Participant 1 was highly responsive to the speed instruction and performed the fast squats with genuinely explosive rotational power, while maintaining a controlled and measured pace during the slow condition.

Participant 2 shows a moderate slow squat peak angular velocity with a noticeable increase during fast squats, and a reasonable reduction in duration. The difference between peak and minimum angular velocity is consistent across both conditions, suggesting a biomechanically stable squat pattern with good deceleration control at the bottom of the movement in both conditions.

Participant 3 displays relatively average values across both conditions but shows one of the longer slow squat durations in the group, indicating a particularly controlled and deliberate pace during the slow condition. The transition to fast squats brings the duration closer to the group average, suggesting that this participant responded well to the speed instruction.

Participant 4 shows a moderate and consistent pattern in both conditions. The peak and minimum angular velocity values are well-balanced, indicating smooth and controlled thigh rotation throughout the squat cycle. The reduction in duration from slow to fast is clear and proportional, reflecting a biomechanically clean execution of both conditions.

Participant 6 has some of the lowest peak angular velocity values in the group for both conditions, suggesting that this participant performed squats with less rotational speed overall. The slow squat duration is close to the group average, but the fast squat duration is also relatively short, indicating that despite lower peak velocities the movement tempo did increase as instructed.

Participant 7 stands out as a notable case. During the slow squat condition, Participant 7 already shows one of the highest peak angular velocities in the entire group, and the repetition duration is among the shortest for the slow condition. This strongly suggests that Participant 7 did not fully comply with the slow squat instruction and was already performing the slow squats at a considerably faster pace than intended. From a biomechanical standpoint, the thigh was rotating rapidly even during what should have been a controlled, low-velocity movement. The fast squat values then show only a modest further increase in peak velocity and duration change, consistent with the interpretation that the slow and fast conditions were not clearly differentiated by this participant.

Participant 8 shows the lowest peak angular velocity values in the group for both conditions, indicating overall lower rotational speed throughout the task. The slow squat duration is close to the group average. This participant appears to have performed both conditions at a consistently lower intensity, possibly reflecting a more cautious or restricted movement style.

Participant 9 presents a similar pattern to Participant 7. The slow squat peak angular velocity is among the highest in the group, and the repetition duration for slow squats is the shortest of all participants, indicating very fast execution during the slow condition. This is a clear biomechanical indicator that Participant 9 also performed fast-paced squats during the slow squat phase, not adhering to the slow tempo instruction. The fast squat values then show a further increase in peak angular velocity and a further reduction in duration, but the contrast between conditions is smaller than for participants who correctly followed the protocol. This non-compliance means that for Participants 7 and 9, the intended comparison between slow and fast squat conditions is less valid, and their data should be interpreted with caution when drawing group-level conclusions about the effect of movement speed.

Participant 10 shows notably high peak angular velocities in both conditions, with one of the highest fast squat peak values in the entire group. The very short fast squat duration confirms highly explosive execution. The slow squat duration is moderate, suggesting reasonable adherence to the slow condition, making the contrast between conditions one of the most pronounced in the group.

Participant 11 shows consistent and moderate values across both conditions with a clear increase in peak velocity and reduction in duration from slow to fast squats, representing a typical and expected response to the speed instruction. This participant can be considered representative of the average group behavior.

Overall biomechanical interpretation

Taken together, the gyroscope results demonstrate that movement speed has a direct and measurable effect on thigh angular kinematics during squatting. The increase in peak angular velocity from slow to fast squats reflects greater rotational power demand at the hip and knee, requiring more rapid muscle force production and higher neuromuscular activation. The reduction in repetition duration confirms that the entire squat cycle — descent, bottom position, and ascent — was completed more rapidly, reducing time under tension. The inter-individual variability in these values reflects genuine differences in movement strategy, physical capacity, and protocol compliance, and highlights the importance of considering individual-level data when interpreting group results in biomechanical studies.

5. EMG analysis

EMG signals from each participant were processed. First, all raw EMG signals were plotted. These raw signals already contain a bandpass filter. Below in figure 6 top panel is an example of the raw EMG signal. All these signals are from participant number 1. Between participants, there were differences between the EMG signals. Some participants had very small EMG signals while others had very strong ones. Some signals had more noise than others.

Next, the EMG signal was rectified, which means that all values are converted to absolute values. This plot can be seen in the middle panel. Next an envelope of the signal was created, and this is presented in the third panel.

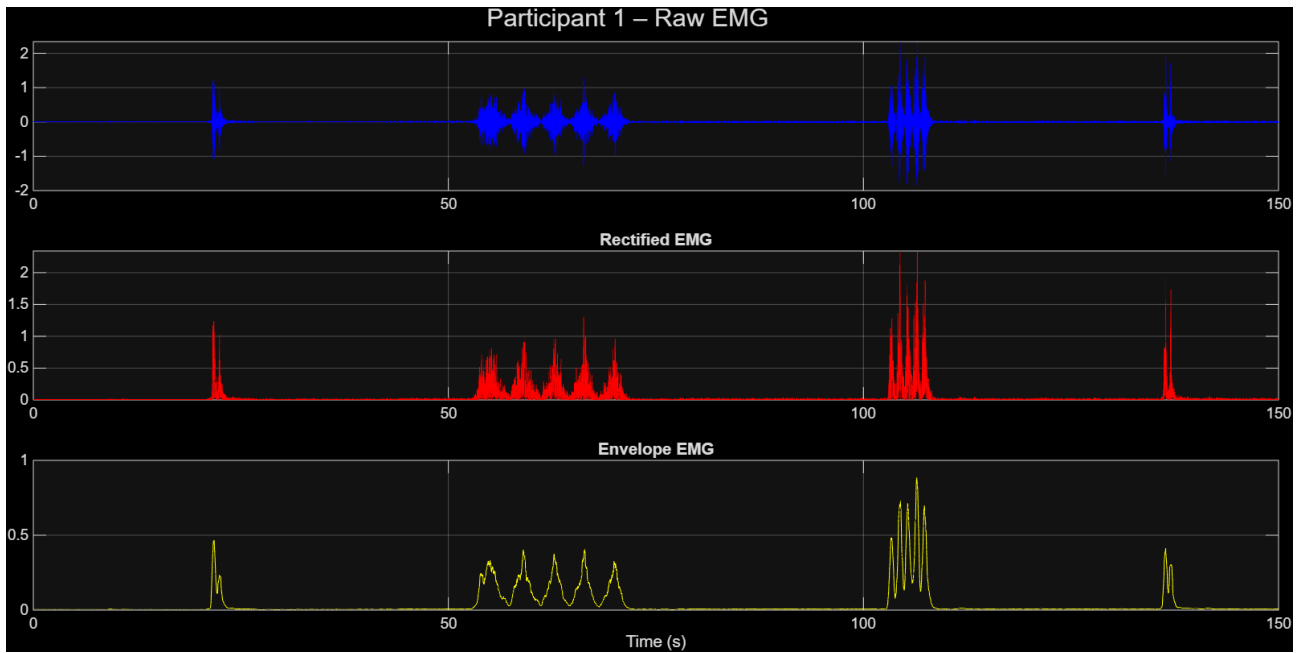


Figure 6: Participant 1 – EMG analysis

Caption: Top panel is raw EMG signal only with bandpass filter. Middle panel is rectified EMG signal, and third panel is an envelope of the EMG signal. As an example, results from participant number 1 are presented here.

From the figure it can be seen that slow squats give wider peaks than fast squats. This was the case with all participants. First and last peaks are caused by jumps at the beginning and at the end of the measurement. Fast squats have higher and stronger peaks than slow squats. This difference between peaks can be expected since fast squats have sharper and stronger movement than slow ones.

Next, RMS, which is root mean square, and iEMG which is integrated EMG were calculated. Results from these calculations are presented in figure 7.

```
Participant 1:
SLOW: RMS=0.1634 iEMG=2.9199
FAST: RMS=0.2868 iEMG=2.0959
Participant 2:
SLOW: RMS=0.1575 iEMG=2.3207
FAST: RMS=0.0119 iEMG=0.1812
Participant 3:
SLOW: RMS=0.2431 iEMG=4.8659
FAST: RMS=0.3370 iEMG=3.5376
Participant 4:
SLOW: RMS=0.2247 iEMG=4.1396
FAST: RMS=0.2873 iEMG=2.7925
Participant 5:
SLOW: RMS=0.1716 iEMG=3.5411
FAST: RMS=0.1305 iEMG=0.7065
Participant 6:
SLOW: RMS=0.1632 iEMG=3.1491
FAST: RMS=0.1807 iEMG=1.2485
Participant 7:
SLOW: RMS=0.0768 iEMG=0.9311
FAST: RMS=0.0025 iEMG=0.0412
Participant 8:
SLOW: RMS=0.1723 iEMG=2.7758
FAST: RMS=0.1863 iEMG=1.5000
Participant 9:
SLOW: RMS=0.1012 iEMG=1.3579
FAST: RMS=0.0171 iEMG=0.2699
Participant 10:
SLOW: RMS=0.1093 iEMG=1.6054
FAST: RMS=0.1463 iEMG=0.7761
```

Figure 7: Results from RMS and iEMG calculations for each participant.

RMS is average amplitude of the signal, and it also shows muscle activation. Results show that the RMS value differs between slow and fast squats and between participants. Some participants have higher values in slow squats and others in fast squats. It would be expected to see higher values with fast squats since amplitude of the signal affects the results. In slow squats, muscle activation can be higher, which would cause that value to be higher. Errors during the measurement cannot be excluded, and those might affect results.

iEMG shows total muscle activation over time. It includes signal amplitude and duration. During slow squats, the same muscles are activated for a longer time. During fast squats muscles are activated with higher energy but for a shorter time. Based on this, it would be expected that slow squats have higher iEMG than fast squats. The results show that with each participant the slow squat has higher iEMG than the fast squat.

RMS values were not consistent between participants, and the results between participants do not indicate the same things. Slow and fast squats both cause higher value. This indicates that RMS results are not reliable, or it is not suitable for this case. iEMG results show that with each participant slow squats cause higher value, which means that slow squats cause higher total muscle activation, which is expected.

6. Results

Table II. Results from each participant and variable with group mean and standard deviation.

Participants	Slow Squats					Fast squats				
	Peak Angular velocity	Minimum Angular velocity	Duration	RMS	iEMG	Peak Angular Velocity	Minimum Angular Velocity	Duration	RMS	iEMG
1	69.95	54.24	20.13	0.1634	2.9199	241.79	132.46	5.64	0.2868	2.0959
2	100.96	86.52	15.19	0.1575	2.3207	142.79	130.29	9.64	0.0119	0.1812
3	74.49	59.34	25.18	0.2431	4.8659	127.48	114.77	11.61	0.3370	3.5376
4	64.17	75.77	18.86	0.2247	4.1396	114.23	102.10	12.17	0.2873	2.7925
6	60.21	48.15	21.77	0.1716	3.5411	116.96	111.91	8.7	0.1305	0.7065
7	130.87	96.29	9.70	0.1632	3.1491	136.41	117.31	8.6	0.1807	1.2485
8	50.87	40.37	16.86	0.0768	0.9311	75.25	58.47	11.5	0.0025	0.0412
9	128.15	88.67	8.73	0.1723	2.7758	221.05	114.61	7.64	0.1863	1.5000
10	93.33	87.94	12.61	0.1012	1.3579	261.48	215.65	1.31	0.0171	0.2699
11	104.10	80.07	14.69	0.1093	1.6054	142.75	125.79	8.08	0.1463	0.7761
Group mean	87.71	71.73	16.39	0.1583	2.7607	158.02	122.34	8.49	0.1586	1.3150
SD	26.71	17.73	4.75	0.0470	1.1266	58.32	36.91	3.07	0.1152	1.1150

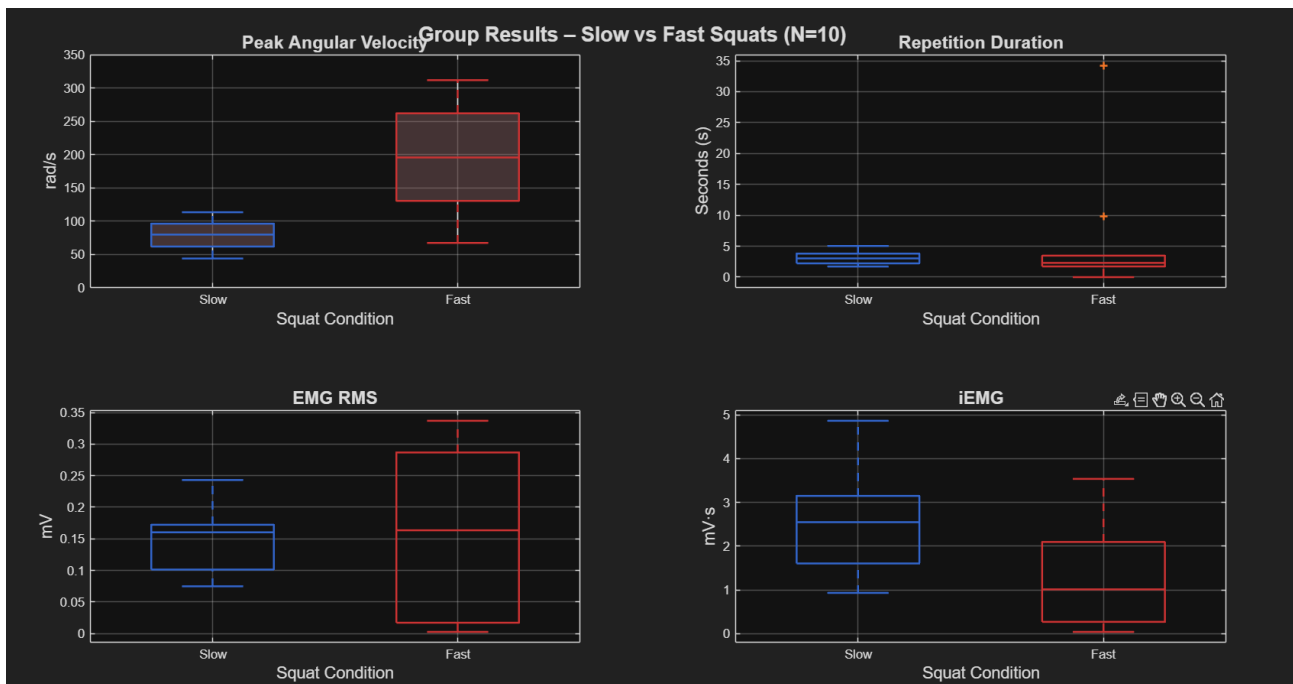


Figure 8: Boxplot of group results.

Caption: Boxplot 1 is peak angular velocity separately for slow and fast squats. Boxplot 2 is repetition duration for both squats. Boxplot 3 is RMS from EMG signal for both squats. Boxplot 4 is iEMG from EMG signal for both squats.

Figure 8 shows a boxplot of group results. First boxplot on top left corner is peak angular velocity. Fast squats have significantly higher overall peak angular velocity than slow squats. Minimum values are almost the same but maximum value in fast squats is much higher. From table 2 it was already seen that standard deviation is almost twice as high with fast squats and this boxplot supports that result.

Second boxplot on top right corner is repetition duration. Slow squats have higher repetition duration but with fast squats it can be seen that there are few outliers from the plot. The median of the plot and results except for outliers fit our results from this experiment. The standard deviation in duration did not have significant variance between slow and fast which can be seen from the plot too.

Third boxplot down left corner is showing RMS from EMG signal. In this plot fast squats have high variability which is showing as a big difference between lower and upper quartile. Lowest and highest value have also quite big difference. Slow squats have smaller variance between lower and upper quartile as well as between highest and lowest value. However, median value is quite close to upper quartile which means that most of the results are closer to upper than lower quartile. Group mean and

standard deviation were close to each other which is also shown in the plot. Main difference is the highest and lowest values and the location of median value in the slow squats plot.

Fourth plot in down right corner is iEMG results. Slow squats have higher median value, lower and upper quartile and minimum and maximum values. Overall, with iEMG results it is clear that slow squats require more total muscle activation over time. This supports our conclusions with this experiment. For iEMG results time affects and even though slow squats might not require more muscle activation, they last longer which then causes higher iEMG values.

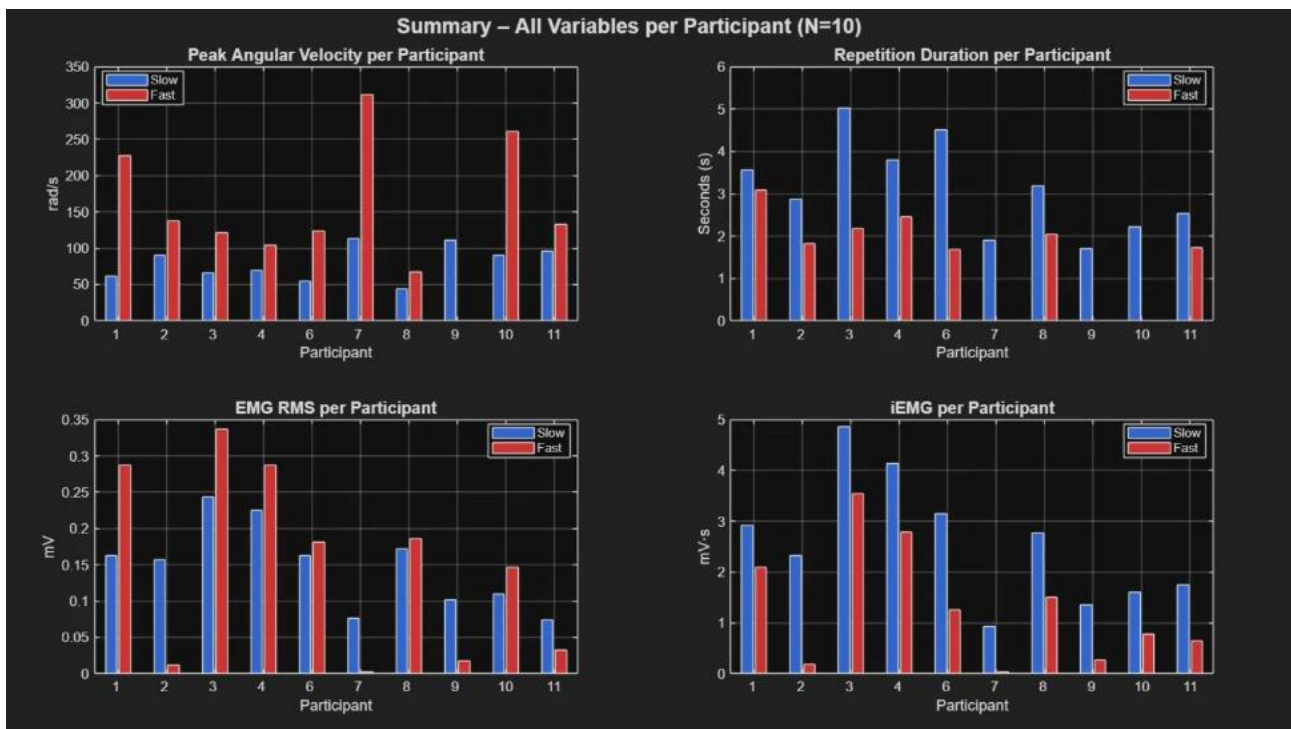


Figure 9: Summary of all variables per participant.

Caption: Plot 1 is peak angular velocity per participant where blue is slow and red is fast squats. Plot 2 is repetition duration in seconds per participant with same colour coding. Plot 3 is RMS from EMG signal per participant and plot 4 is iEMG from EMG signal per participant with same colour coding.

In figure 9 each participant is presented with all variables. This figure demonstrates the difference between participants and between results. Top left corner is peak angular velocity per participant. With participant number 8 it can be seen that there is not much difference between slow and fast squats. This means that the maximum rotational speed of thigh did not change much between squats. Participant 7 on the other hand has a big difference and movement might have been really

explosive. It would be expected to see a difference and this figure also shows that instructions were followed correctly.

Second plot in the right upper corner is repetition duration in seconds per participant. This plot clearly shows that fast squats which are marked with red colour have shorter duration and this is the case with every participant. This indicates that instructions were followed correctly and everyone did squats with different duration. It also shows the variance between participants. Standard deviation in both squats was quite small but still there are differences. Because each participant is different it is expected that repetition duration is not the same for everyone.

Third plot in the down left corner is presenting RMS from EMG signal per participant. This shows the variance between participants. Some have higher RMS during fast squats and others during slow ones. Standard deviation in RMS was small. RMS is affected by many factors which then cause variance. This also demonstrates that physics of each participant is different.

Fourth plot in down right corner shows an iEMG signal per participant. It is clear that every participant has higher iEMG values during slow squats. This is lined with our other iEMG results which indicate that total muscle activation over time is higher during slow squats.

Overall, from figure 9 it can be seen that variance between participants is high. This variance is caused by every participant being an individual human. Factors such as physics, exercise history, flexibility and injuries affect the results and cause variability. These variabilities can be seen in the results, but the overall quality of the results seems to be good.

7. Interpretation

Results from this signal analysis show that slower squats take more time and require more muscle activation. From gyroscope sensors the difference between slow and fast squats could be clearly seen. Fast squats have higher angular velocity and peaks caused by these are sharper than in slow squats. It was found that gyroscope gives better signal than accelerometer. This is because gyroscope measures angular velocity which is rotation. Squats are rotational movements which are then detected better with gyroscope.

From EMG signals it could be seen that fast squats have higher EMG signal than slow squats. EMG measures muscle activation level and slow squats require less muscle activation but for a longer time. Movement is more controlled and stable during the whole squat which requires muscle activation during the whole movement. During fast squats muscle activation is higher but required for a shorter time. This causes high but narrow peaks.

There was a variability between participants. Two excluded participants had signals that clearly were not suitable for use. Between the rest of the participants, there was variance too. Some participants had a lot more noise in the data than others. The shape of the data from the same sensor was different between participants. The length of the squats as well as values that were obtained from the squats were different in each dataset. Each person has different flexibility, movement range, and muscle forces. Injuries or any limitations to movement will affect this kind of measurement and results. Our data does not include any mentions about possible restrictions, so the effect of these cannot be excluded completely.

In slow squats highest standard deviation was at peak angular velocity. This means maximum rotational speed of thigh during the squat. Each participant has different flexibility, amount of muscle force and movement even though everyone was doing squats. These are the factors that most likely cause highest standard deviation. In fast squats maximum peak velocity also caused the highest standard deviation. Compared to slow squats the standard deviation is almost twice as high in fast squats.

Minimum angular velocity was also over twice as high in fast squats than in slow squats. This is the slowest point of the squat which is the transition point at the bottom. During fast squats movement is faster and sharper which explains why the mean value of minimum peak velocity is higher during fast squats. Time that body is in the transition phase is shorter and body does not stop at the bottom. During slow squats the body might stop at the bottom which means that the slowest point is really slow.

The duration of the slow squats was twice as high as fast squats. This is how it is supposed to be, and it shows that each participant did the slow squats as they were supposed to be done. Standard deviations between these two are close together. Because only duration is measured, there are not many variables that can affect the duration.

RMS was almost the same between squats, but standard deviation was little bit higher during fast

squats. Since RMS shows average signal amplitude and muscle activation, there are many factors affecting the results. It would be expected that RMS will be higher during fast squats, but our measurements show that it is not. Neuromuscularity might also affect RMS results, and it is different between every participant. For example, previous exercise history and condition of muscles are factors that affect the muscle activation needed for squats. This also explains why RMS results are quite close to each other and why there is not big difference between fast and slow.

iEMG was higher during slow squats and so was standard deviation. iEMG represents total muscle activation over time, it is expected to see higher results during slow squats and our results support that. Slow squats take longer time, which causes higher total muscle activation for a longer time.

Overall, the main difference between slow and fast squats is different kinds of signal for gyroscope and EMG sensors and differences in main variables. Duration changes as well as maximum and minimum peak angular velocity. These are completely caused by differences in the movement between squats. Total muscle activation changes based on the duration of the squat. Our data analysis does not show differences in RMS.

Based on our data, the main difference between slow and fast squats is the speed of thigh rotation and the total duration of the movement, which together drive all other observed differences. Fast squats produce significantly higher peak angular velocities and shorter repetition durations, reflecting greater rotational power output and more explosive neuromuscular demand at the hip and knee joints. While RMS amplitude remained similar between conditions, iEMG was higher during slow squats due to the longer time under tension, confirming that slow squats require sustained muscle activation over a greater duration rather than a higher instantaneous force output. These findings demonstrate that movement speed fundamentally changes the mechanical and neuromuscular strategy of the squat, and that both gyroscope and EMG signals are necessary to capture the full picture of this difference.

8. Group challenge

Selected signals: Gyroscope and EMG combined

After comparing all three available signals — thigh accelerometer, gyroscope, and EMG — our conclusion is that neither a single signal alone provides the most complete picture of squat performance. The gyroscope and EMG together form the most reliable and informative combination for analyzing squat performance, as they capture two fundamentally different but complementary aspects of movement: kinematics and neuromuscular activation. Each signal answers a different

biomechanical question, and together they allow a much deeper interpretation than either could provide independently.

Why the gyroscope is the primary kinematic signal

The thigh gyroscope is the most reliable single signal for detecting and quantifying squat movement. It directly measures the angular velocity of the thigh segment, which reflects the rotational motion occurring at the hip and knee joints during each squat repetition. Because the gyroscope is insensitive to gravitational orientation, it produces a clean, consistent signal that clearly captures the descent and ascent phases of each squat as a distinct biphasic pattern. This makes it highly suitable for automated repetition detection, calculation of peak angular velocity, and measurement of repetition duration. The gyroscope also clearly distinguishes between slow and fast squat conditions based on peak amplitude and timing alone, making it directly useful for performance comparison across participants and conditions.

Why EMG is essential as a complementary signal

While the gyroscope describes how the body moves, EMG describes how the muscles are working to produce that movement. This distinction is critical because two participants can produce identical kinematic patterns while using very different levels of muscular effort. EMG captures the neuromuscular activation of the muscle, reflecting the intensity and timing of muscle recruitment throughout the squat cycle. This makes it an indispensable signal for understanding the physiological demand of the movement, not just its mechanical output.

The value of combining gyroscope and EMG becomes especially clear when examining Participants 9 and 10 in figure 10 and 11. Looking first at their gyroscope signals, both participants produced high peak angular velocities exceeding 200 rad/s during fast squats, indicating similarly explosive kinematic performance.

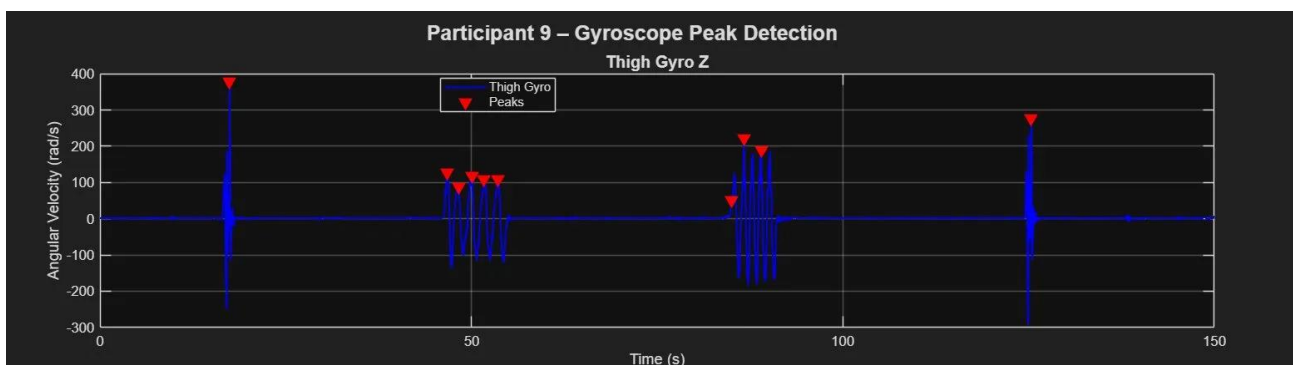


Figure 10: Participant 9 — Gyroscope Peak Detection

Caption: Participant 9 thigh gyroscope showing high-amplitude fast squat peaks exceeding 200 rad/s during the fast squat period.

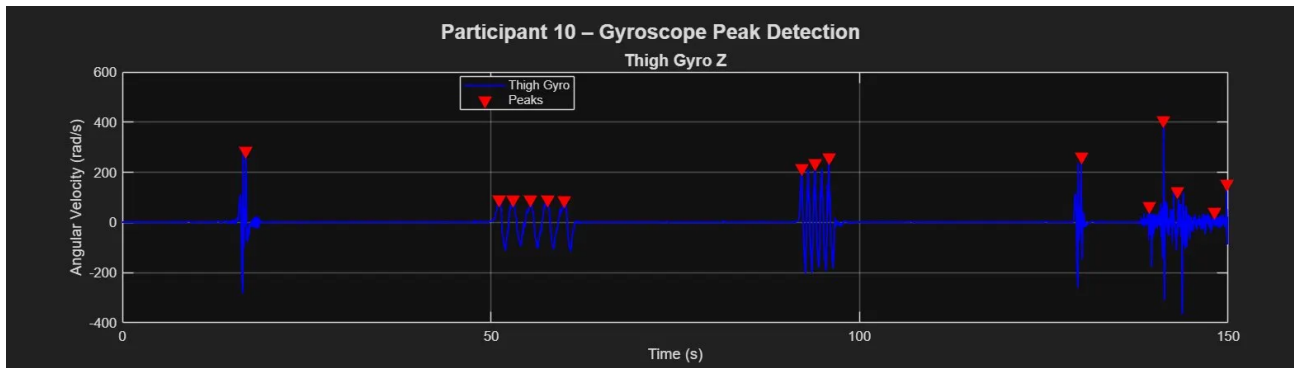


Figure 11: Participant 10 — Gyroscope Peak Detection

Caption: Participant 10 thigh gyroscope showing comparable high-amplitude fast squat peaks, visually similar in magnitude to Participant 9.

However, when their EMG envelopes are examined directly alongside these gyroscope results, a very different picture emerges. Participant 9 in figure 12 shows considerably higher and more sustained EMG activation amplitudes throughout both squat conditions, while Participant 10 in figure 13 achieves comparable or even higher angular velocities with noticeably lower muscle activation levels overall.

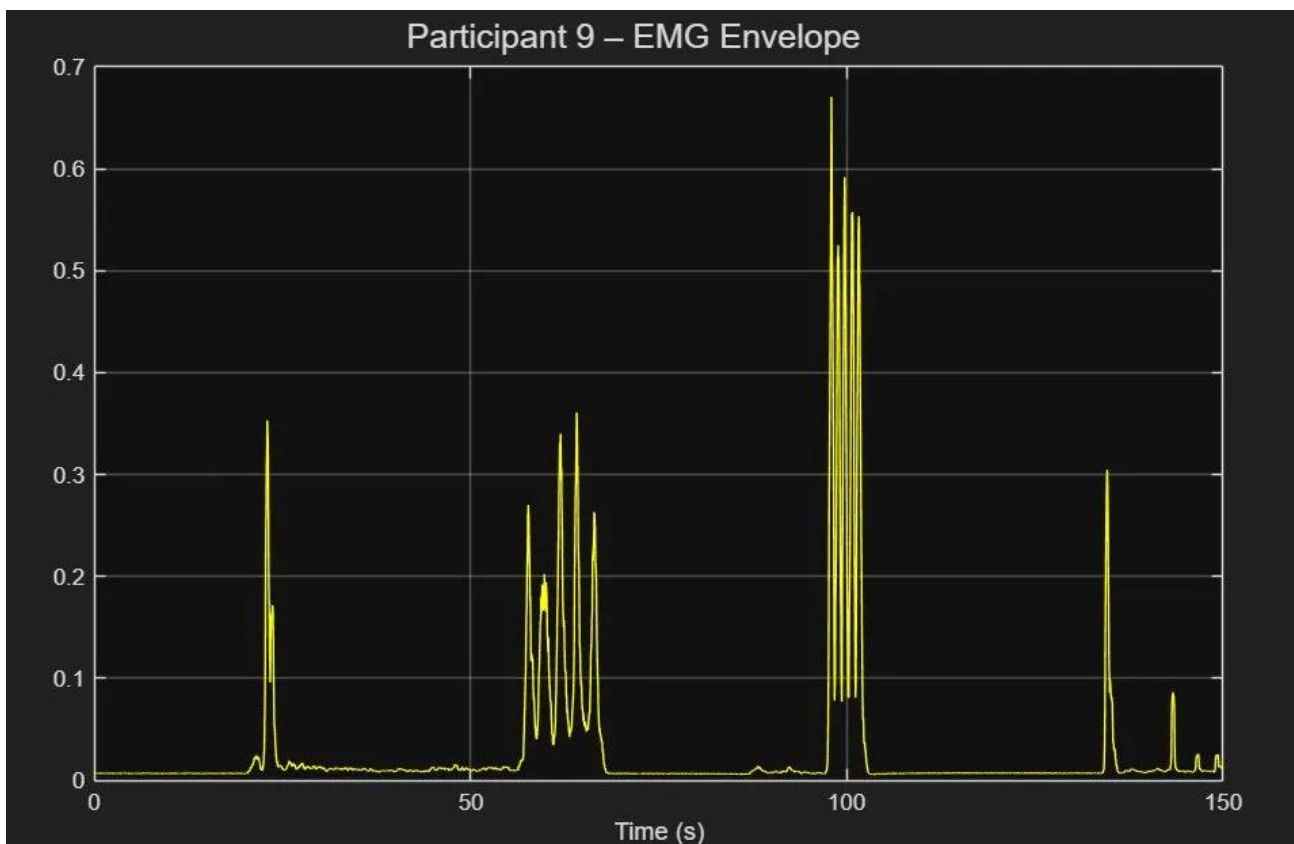


Figure 12: Participant 9 — EMG Envelope

Caption: Participant 9 EMG envelope showing high activation amplitudes during both slow and fast squat periods, with particularly prominent peaks during fast squats around 95–100 seconds.

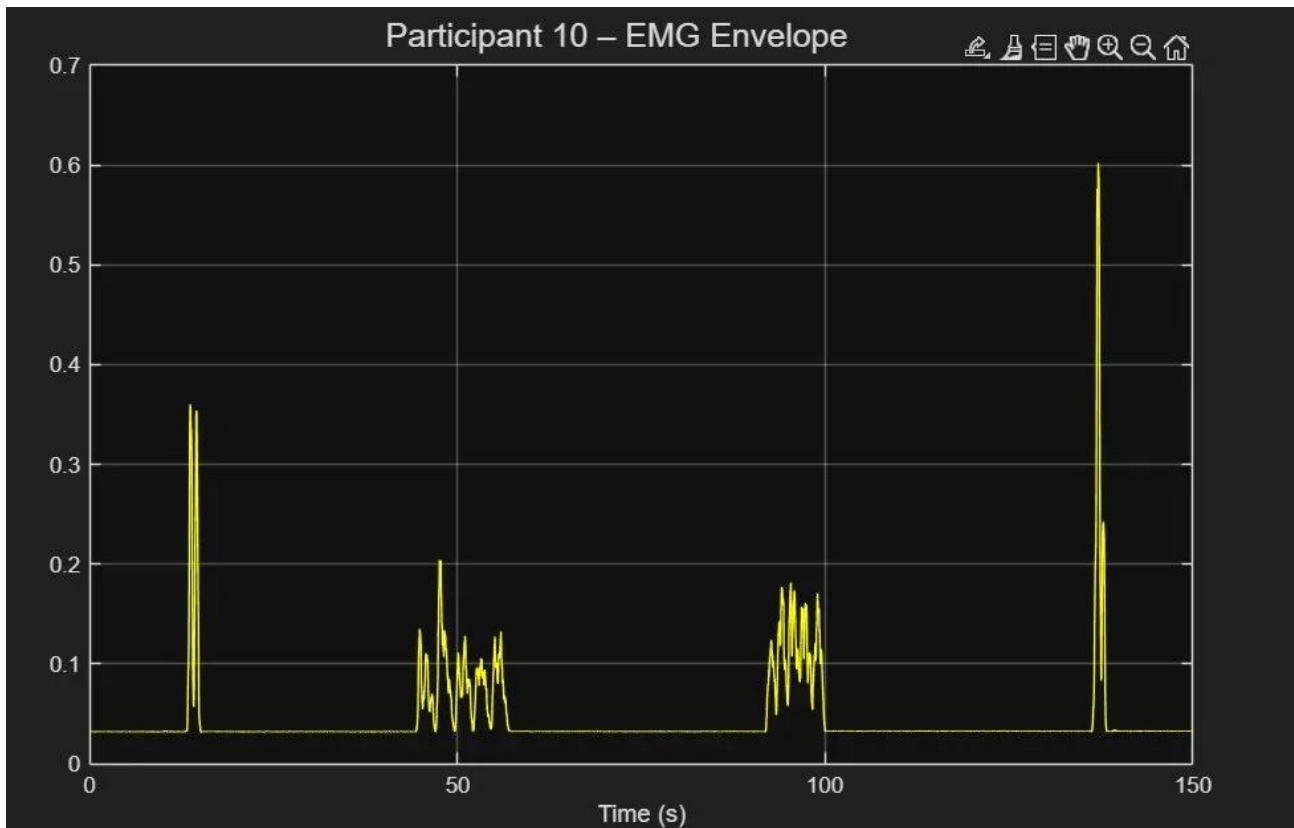


Figure 13: Participant 10 — EMG Envelope

Caption: Participant 10 EMG envelope showing lower overall activation amplitudes during both squat periods despite producing similarly high angular velocities to Participant 9.

This reveals that Participant 10 achieved comparable explosive performance with less muscular effort, suggesting greater movement efficiency or neuromuscular economy. This kind of insight — the relationship between mechanical output and muscular cost — is completely invisible to the gyroscope alone and can only be revealed by examining both signals together.

Identifying the source of movement restriction

The combination of signals also allows identification of cases where restricted movement is not caused by reduced muscle activity. Participant 8 in figure 14 shows relatively low peak angular velocities in the gyroscope signal during both conditions compared to the group average, which might initially suggest low physical effort or limited movement capacity.

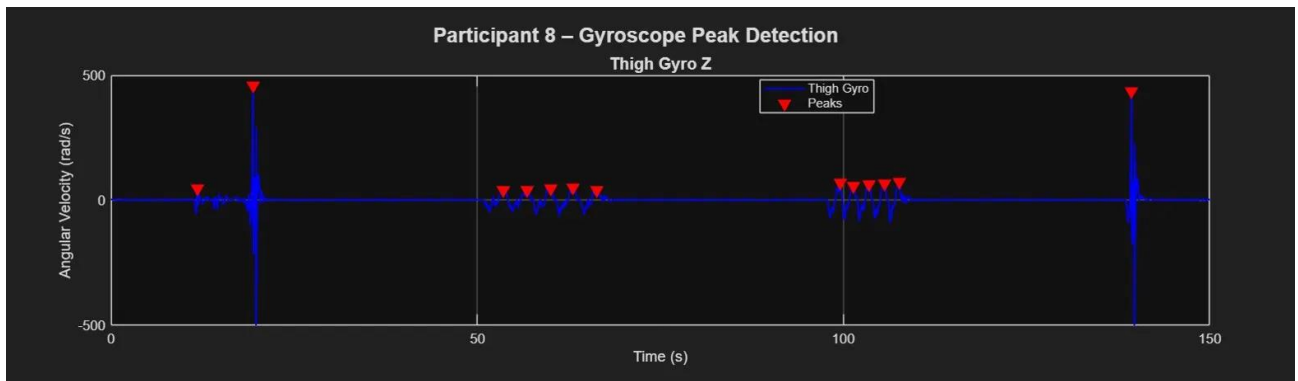


Figure 14: Participant 8 — Gyroscope Peak Detection

Caption: Participant 8 thigh gyroscope showing low-amplitude squat peaks during both slow and fast conditions compared to the group, suggesting restricted rotational movement of the thigh.

However, when Participant 8's EMG envelope in figure 15 is examined immediately alongside this, the picture changes completely.

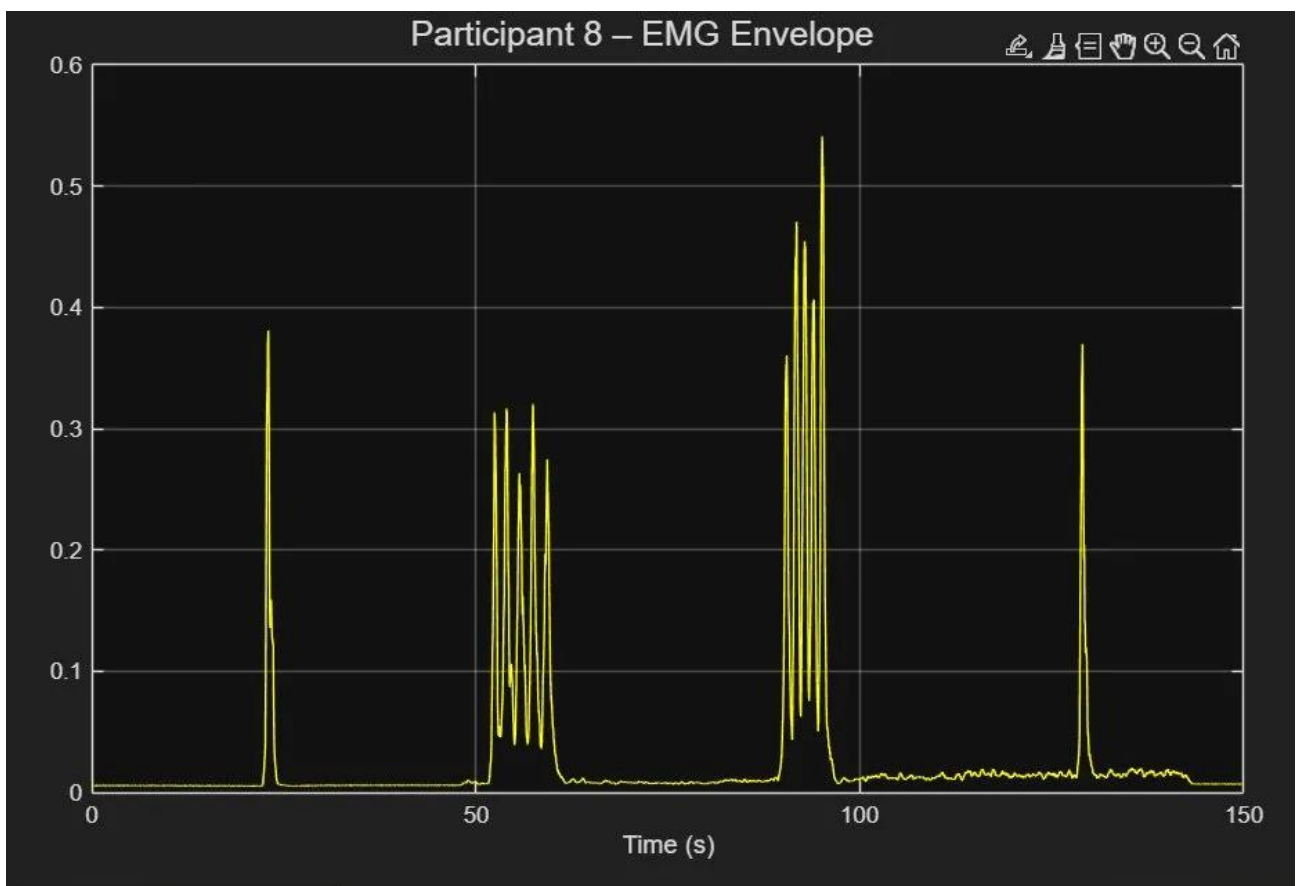


Figure 15: Participant 8 — EMG Envelope

Caption: Participant 8 EMG envelope showing normal and relatively high activation amplitudes during both squat periods, particularly elevated during the fast squat phase around 90–100 seconds, despite the restricted kinematics seen in the gyroscope.

The muscles are clearly working normally and even vigorously, yet the thigh rotation remains restricted. This demonstrates that the movement restriction visible in the gyroscope is not due to insufficient muscle activation but may instead reflect other factors such as joint stiffness, individual body mechanics, or movement strategy. Without the EMG signal this distinction would be impossible to make, and the interpretation of Participant 8's performance would be incomplete or misleading.

Why is the thigh accelerometer less suitable

The thigh accelerometer measures linear acceleration along the Z-axis, which contains a mixture of dynamic movement-related acceleration and the constant gravitational component. During squatting, the changing orientation of the thigh relative to gravity causes continuous variation in the gravitational projection onto the Z-axis, which overlaps with and partially masks the actual movement signal. This makes the accelerometer signal more complex to interpret and more difficult to use for reliable repetition detection without additional preprocessing. It is therefore considered a supplementary rather than primary signal for squat performance analysis.

Conclusion

The gyroscope provides the most reliable kinematic information for detecting repetitions, quantifying movement speed, and comparing conditions, while EMG adds the essential neuromuscular dimension that explains how muscles produce and control that movement. Together, as demonstrated through Participants 8, 9, and 10, they enable a complete biomechanical interpretation of squat performance — one that can identify not only what the movement looks like, but how efficiently and effectively the neuromuscular system is driving it.

9. Product concept

The problem

Weightlifting movements such as the clean, snatch, and deadlift require precise coordination of lower limb muscle activation and joint kinematics, particularly involving the thigh and calf segments. In competitive and recreational weightlifting, improper technique — such as asymmetric force generation, incorrect movement tempo, or insufficient muscle activation during the critical lifting phase — is a leading cause of both suboptimal performance and injury. Currently, most athletes rely on coach observation or video analysis to identify technical errors, which is subjective, not always available in real time, and provide no quantitative feedback on the underlying neuromuscular and kinematic patterns driving the movement. There is a clear need for an objective, data-driven tool that

can monitor and evaluate weightlifting technique continuously and provide immediate actionable feedback to both athletes and coaches.

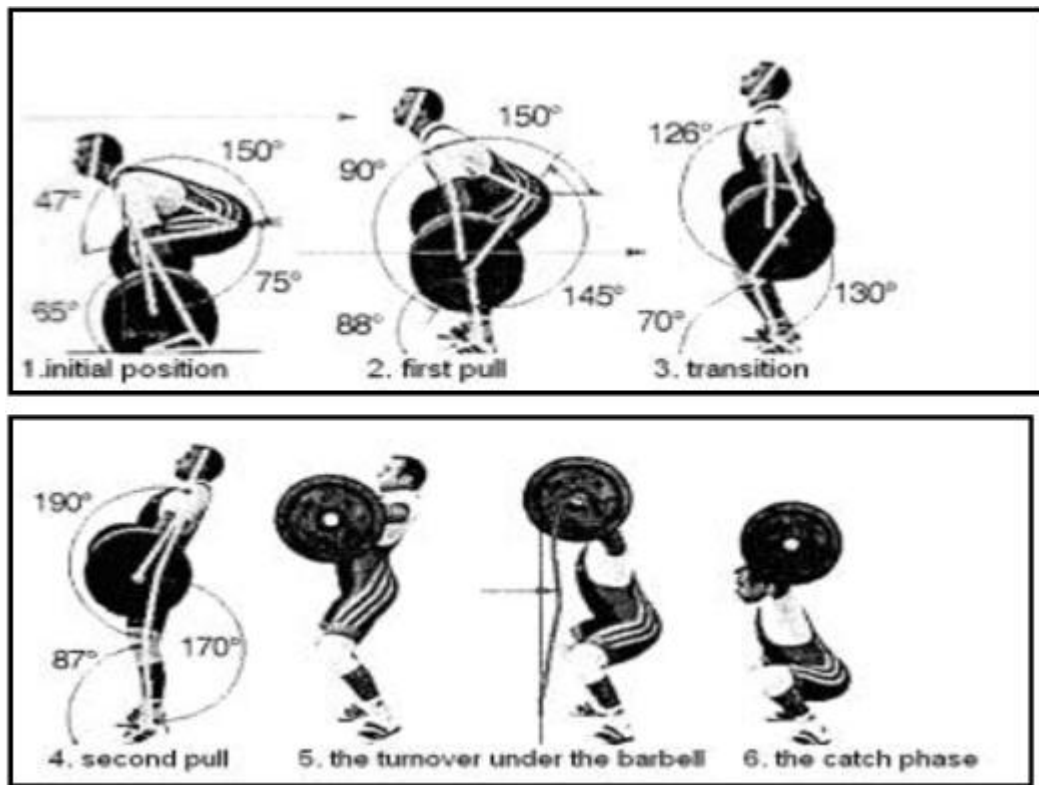


Figure 16: Gaits in weightlifting (Himawan et al., 2018)

The product

The proposed product is a wearable sensor-based mobile application designed specifically for weightlifting training and performance monitoring. The system would consist of two lightweight IMU sensors — one attached to the thigh and one to the calf — combined with surface EMG electrodes placed on the quadriceps or hamstrings, all wirelessly connected to a smartphone or tablet application. During a lifting session, the application would continuously record gyroscope signals from both segments and EMG activation from the target muscle group. Using signal processing methods similar to those applied in this study — peak detection, angular velocity calculation, repetition segmentation, and EMG envelope extraction — the app would automatically analyze each repetition in real time and present the results to the user in an intuitive visual dashboard.

The application would be able to detect each repetition automatically, calculate the angular velocity profile of the thigh and calf throughout the lift, identify the timing and magnitude of peak muscle activation, and compare the movement pattern against a reference template of correct technique. If the system detects deviations — such as a peak angular velocity that is too low indicating insufficient

drive from the legs, asymmetric timing between segments, or inadequate muscle activation during the key power phase — it would generate an immediate alert or visual feedback on the screen. Over multiple sessions, the app would track progress by storing historical data and displaying trends in peak angular velocity, repetition duration, and EMG amplitude, allowing athletes to monitor improvement objectively over time.

An additional simulation mode could allow athletes to train specific aspects of their technique without a loaded barbell, using the sensor data alone to evaluate movement quality and compare it against their own best performances or against group benchmarks derived from a database of experienced lifters.

Target users

The primary target users are competitive and recreational weightlifters at all levels, from beginners learning correct technique to advanced athletes optimizing performance. The product would also be highly relevant for strength and conditioning coaches who work with weightlifters, powerlifters, or any athletes whose sport requires explosive lower limb force production such as sprinters, jumpers, or rugby players. A secondary user group would be physiotherapists and sports medicine professionals working with athletes returning from lower limb injuries, as the system could be used to monitor movement quality and muscle activation symmetry during rehabilitation and return-to-sport assessment.

Why this would be useful

The core benefit of this product is that it transforms subjective technique assessment into objective, quantitative, real-time feedback. An athlete can immediately understand whether their drive phase was powerful enough, whether their movement tempo was consistent across repetitions, and whether their muscles are activating at the right moment and with sufficient intensity. This removes the guesswork from training, reduces reliance on constant coach supervision, and allows athletes to self-correct and self-monitor independently during solo training sessions. The historical tracking feature also provides motivation by making progress visible and measurable. From a safety perspective, the system could flag repetitions where movement patterns deviate significantly from the norm, potentially indicating early signs of fatigue or compensatory movement that could lead to injury if continued.

Signal selection and justification

The product would primarily use the thigh gyroscope signal combined with the EMG signal, for the same reasons established throughout this analysis. The thigh gyroscope provides the most direct and reliable measure of lower limb kinematics during the lift. In weightlifting, the explosive upward drive from the legs produces a rapid and powerful rotation of the thigh, which is captured cleanly by the gyroscope as a high-amplitude peak. The timing, magnitude, and shape of this peak directly reflect the quality and power of the leg drive phase — the most critical mechanical component of any weightlifting movement. The gyroscope is also unaffected by gravitational orientation changes, making it robust across the full range of motion involved in lifting.

The EMG signal adds the neuromuscular dimension that the gyroscope alone cannot provide. As demonstrated in this study, two athletes can produce similar angular velocities with very different levels of muscle activation, reflecting differences in efficiency and technique. In weightlifting, monitoring EMG activation allows the system to determine whether the correct muscles are being recruited at the correct moment, and whether the athlete is generating force efficiently or compensating with suboptimal muscle coordination. The calf gyroscope provides additional supporting information about lower leg contribution to the lift, which is particularly relevant in movements where ankle extension plays a significant role in force transmission.

The thigh accelerometer would serve as a supplementary signal only, providing additional context about linear segment motion but not used as a primary performance metric due to the gravitational confound discussed earlier in this report.

10. Feedback

The computer classes were well structured and the tasks followed a logical progression from individual to group level analysis. The lab sessions were hands-on and working with real measurement data made the learning more meaningful. Having MATLAB as the main tool worked well as it allowed direct implementation of the signal processing steps covered in the course.

The most challenging part was the repetition detection using peak detection, as tuning the parameters required a lot of trial and error. Understanding the difference between RMS and iEMG and when each metric is more appropriate was also initially unclear but became clearer through the analysis.

Almost spent 2 days for coding, Day for analysing and reporting two days group work and product concepts.

11. Use of AI

During EMG analysis AI was used to detect errors in the code. Used for correcting grammar and got guide for structured report.

12. References

- Himawan, M.K.N., Rilastia, D., Syafei, M., Nugroho, R. and Budihardjo, B. (2018) 'Biomechanical Analysis of Snatch Technique in Conjunction to Kinematic Motion of Olympic Weightlifters', in *Proceedings of the International Seminar on Public Health and Education 2018 (ISPHE 2018)*. Semarang, Indonesia: Atlantis Press, pp. 132–137. Available at: <https://doi.org/10.2991/isphe-18.2018.30>